

What do we estimate when we estimate excess mortality?

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1. Framing the question
2. What is your estimand?
3. The modeled quantity
4. Modeling the counterfactual
5. Data & denominators
6. Synthesis

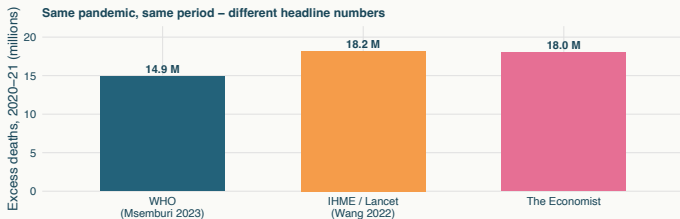
Framing the question

What is excess mortality?

- Excess mortality became one of the most widely-used summary metrics during COVID-19.¹
- It is often presented as if it were a single, well-defined number:

$$\text{excess deaths} = \text{observed deaths} - \text{expected deaths}.$$

- This simplicity hides substantive choices — about assumptions, data, **estimands**, and models — and different groups produced very different COVID-era estimates, not from statistical noise alone.
- **Goal of this talk:** unpack what we are actually estimating, and where the choices live.



¹ Msemburi et al., 2023; Wang et al., 2022a; Karlinsky and Kobak, 2021

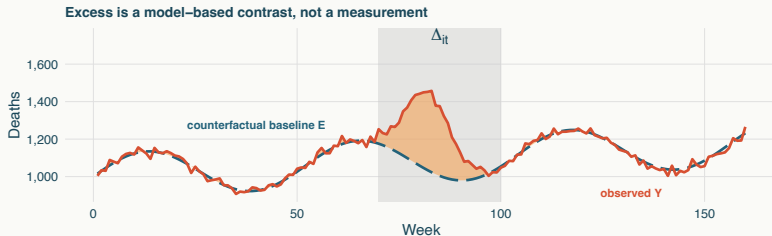
What is excess mortality?

A counterfactual contrast:

$$\Delta_{it} = Y_{it} - E_{it}$$

where, for stratum i and time t ,

- Y_{it} is observed mortality (a count, a rate, a life-table summary, ...),
- E_{it} is mortality in a counterfactual world without the shock of interest.



What is excess mortality?

Key features:

- Y_{it} is, in principle, observable — but almost always imperfectly.²
- E_{it} is *never observed*; it must be modeled.
- Excess mortality estimation is fundamentally a problem of **counterfactual inference**, not measurement.³

We do not directly measure excess mortality.
We *estimate* an unobservable contrast.

Everything that follows in this talk — modeling target, assumptions, baseline, window, scale — is a choice that defines *which* counterfactual contrast we are estimating.

² Mikkelsen et al., 2015; Adair and Lopez, 2018

³ Wakefield and Knutson, 2025

Excess mortality is not one question

The same formula answers several distinct questions:

Burden

“How many additional people died?”

Risk

“How much higher was mortality risk?”

Disruption

“How abnormal was mortality relative to expectation?”

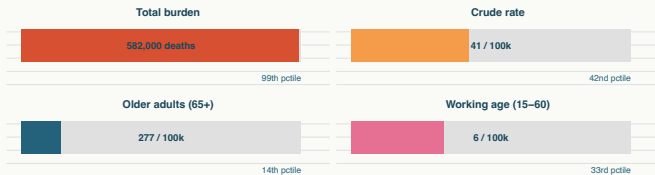
Causal effect

“What changed *because* the pandemic occurred?”

Example — China, 2022: #2 in the world by total excess deaths, yet only mid-pack by crude rate and low by age-specific rates. None is wrong; they answer different questions.

China, 2022: one country–year, four different answers

Bar = China's rank among 195 countries (WHO estimates)

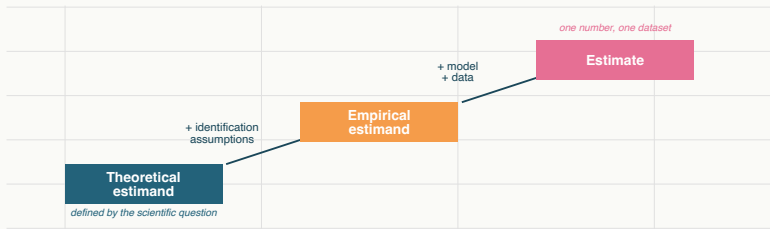


What is your estimand?

The estimand comes first

Be explicit about the ***theoretical estimand*** before specifying the model.⁴

From question to number: each arrow adds an assumption



⁴ Lundberg et al., 2021

What the theoretical estimand requires us to choose

For excess mortality, defining the theoretical estimand requires at least six choices:

- **Population:** adults? a vulnerable subgroup?
- **Time horizon:** weekly, annual, “full epidemic”?
- **Geographic unit:** national, subnational, small-area?
- **Counterfactual:** pandemic absent entirely? absent but health system unchanged?
- **Contrast:** difference, ratio, log-ratio?
- **Summary:** deaths, rates, e_0 , YLL, P-score?

Different excess mortality papers estimate *different things*, even when they all print the word “excess.” The rest of this talk walks through these six axes.

The modeled quantity

Modeling target: excess counts

Estimand: $\Delta_{it}^D = D_{it}^{\text{obs}} - D_{it}^{\text{cf}}$.

Typical model: a count regression for D_{it} ,

$$D_{it} \sim \text{NegBin}(\mu_{it}, \phi), \quad \log \mu_{it} = \alpha_i + s_t + f(\text{season}_t) + \mathbf{x}_{it}^\top \beta.$$

Interpretation: “How many *more* deaths occurred?”

Strengths

- additive, intuitive,
- links directly to resource planning,
- the public communicates in counts.

Weaknesses

- counts conflate *risk* with *population size, structure, and migration*,
- cross-country comparisons mislead without standardization.⁵

COVID example: the headline “~ 15 million global excess deaths” framings are count-based.⁶

⁵ Nepomuceno et al., 2022

⁶ Msemburi et al., 2023; Wang et al., 2022a

Estimand: $\Delta_{it}^{\lambda} = \lambda_{it}^{\text{obs}} - \lambda_{it}^{\text{cf}}$. Same count model, with a denominator and offset:

$$D_{it} \sim \text{NegBin}(P_{it} \lambda_{it}, \phi), \quad \log \mu_{it} = \log P_{it} + \alpha_i + s_t + \dots$$

Interpretation: “How much higher was *mortality risk*?”

Strengths

- population-normalized;
age-standardizable,
- epidemiologically interpretable.

Caveats

- requires a credible P_{it} ,
- migration / census drift can dominate.⁷

⁷ Nepomuceno et al., 2022

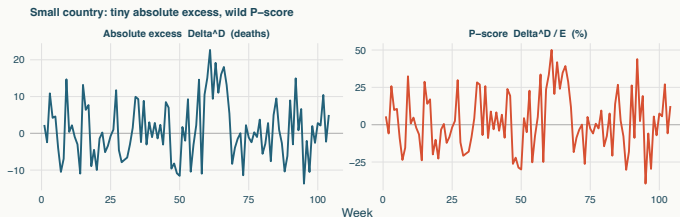
Modeling target: relative excess

Estimand (multiplicative):

$$R_{it}^{\lambda} = \frac{\lambda_{it}^{\text{obs}}}{\lambda_{it}^{\text{cf}}}, \quad \text{or the } P\text{-score: } P_{it} = \frac{D_{it}^{\text{obs}} - D_{it}^{\text{cf}}}{D_{it}^{\text{cf}}}.$$

Interpretation: “How much *above normal*?” — a unitless disruption metric⁸

- Natural for relative disruption (intervention / interrupted-time-series framing); convenient across very different baseline scales.
- **Caveat:** unstable when D_{it}^{cf} is small; a 50% P-score is *not* comparable across high- vs. low-mortality settings.



⁸ Aron et al., 2020; Karlinsky and Kobak, 2021

The modeling target encodes a *mechanism*:

$$\underbrace{D_{it} = D_{it}^{\text{cf}} + \Delta_{it}}_{\text{additive shock}} \quad \text{vs.} \quad \underbrace{D_{it} = D_{it}^{\text{cf}} \cdot R_{it}}_{\text{multiplicative shock}}$$

- **Additive:** the shock *adds* deaths onto the baseline — e.g. a short, well-bounded natural disaster.⁹
- **Multiplicative:** the shock *scales* baseline mortality — e.g. an infection hitting vulnerable/non-vulnerable in proportion to baseline hazard, arguably closer to many COVID-19 settings.¹⁰

⁹ Hajat et al., 2005

¹⁰ Aburto et al., 2022; Kontis et al., 2020

Modeling the counterfactual

Classical / surveillance

- **Serfling-style cyclic regression**¹¹ — trend + harmonic seasonality; needs a clean in-sample “normal” period (influenza-surveillance origins).
- **Farrington / aberration detection** — outbreak-surveillance heritage; short-window flags, conservative on long smooth shocks.
- **Poisson / negative-binomial regression** — covariates, offsets, etc.

Hierarchical & ensemble

- **Bayesian hierarchical / spatiotemporal**¹² — borrow strength across strata.
- **Ensembles / model averaging**¹³ — BMA, stacking, etc.

Each family encodes assumptions about smoothness, stationarity, and what counts as a “normal” period. Two well-fit models can produce reproducibly different Δ_{it} .

¹¹ Serfling, 1963

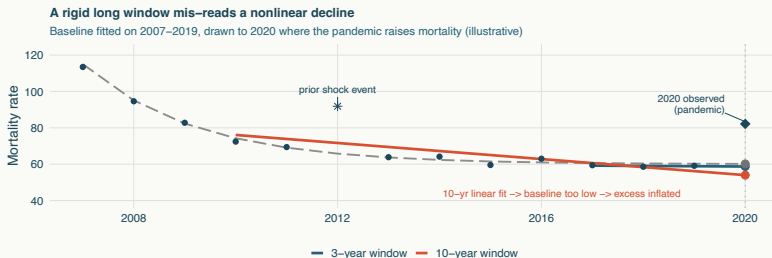
¹² Wakefield and Knutson, 2025; Acosta and Irizarry, 2022; Barnard et al., 2022

¹³ Wang et al., 2022a; Msemburi et al., 2023

Choosing the reference period: what counts as “normal”?

The question: how many pre-shock years should define the baseline?

- **Short (3y):** tracks the recent regime — accurate for stable trends, but noisier and hostage to a single bad year.
- **Long (10y+):** more data — but a *rigid* fit mis-reads a nonlinear decline, and a prior shock event inside the window contaminates the baseline.



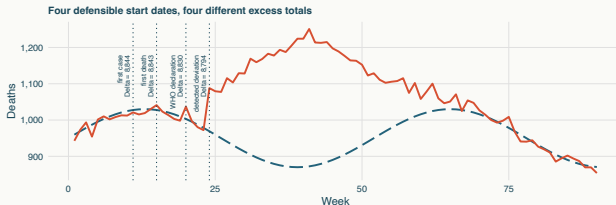
When does the shock start?

The window $[\tau_0, \tau_1]$ defining the “event” is rarely innocuous.

Candidate τ_0 : WHO declaration; first reported case/death; first *detected* deviation (data-driven, but circular); a wave window or fixed calendar bin (e.g. 2020–22).

Two opposite biases:

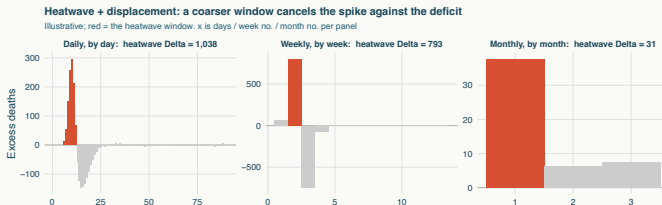
- pre-shock weeks inside the event window $\Rightarrow$ contaminates E_{it} , deflates Δ_{it} ;
- event weeks left in the baseline $\Rightarrow$ inflates E_{it} , deflates Δ_{it} .



Temporal granularity changes the estimand

- **Daily:** resolves acute waves/heatwaves, but noisy with reporting artifacts.
- **Weekly:** the de facto compromise (Eurostat, WMD) — signal vs. noise.
- **Monthly:** smoother and more robust, but can mask short waves.
- **Annual:** communicable and comparable, but obscures waves, seasonality, displacement.

Weekly and annual models are not just different resolutions of one answer: implicit baseline smoothness and shock-window differ, so integrated Δ 's need not coincide.



Negative excess mortality is informative

A negative Δ_{it} is not necessarily a bug. Sources:

Real reductions in mortality

- suppressed influenza circulation (NPIs in 2020–21),
- fewer road traffic fatalities under lockdown,
- reduced air pollution, altered behavior.

Mechanistic redistributions

- mortality displacement after an earlier wave,
- mortality selection: the survivor pool is healthier on average.

Artifacts

- baseline misspecification (over-fit secular trend),
- denominator errors (missing migration),
- incomplete or delayed reporting.

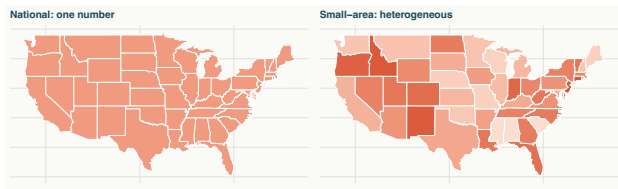
Negative excess is a diagnostic — it tells us the model, the data, or the world is doing something interesting. None of the three is the default explanation.

Spatial granularity changes the estimand

- **National:** stable, but masks within-country heterogeneity.
- **Subnational:** reveals the gradient; loses power in small populations.
- **Small-area:** useful takeaways, but counts turn sparse and noisy.

At small scales the estimand *requires* a smoothing assumption to be identified: independent areas give unstable rates, while smoothing models borrow strength across neighbors¹⁴ — the smoother is *part of the estimand*, not a side technicality.

COVID example: within-country heterogeneity (U.S., Indian, Brazilian subnational units) often *exceeds* between-country differences.¹⁵



¹⁴ Rue and Held, 2005; Rue et al., 2017

¹⁵ Wakefield and Knutson, 2025; Wrigley-Field et al., 2025

Data & denominators

Best case: vital registration

- high-income settings, near-complete VR: could still contain reporting delay, definitional change, late-corrected data
- Death Distribution Methods (GGB, SEG, hybrid) recover completeness from censuses plus a reporting model¹⁶
- Estimand becomes: “true mortality *after* adjusting for under-reporting”

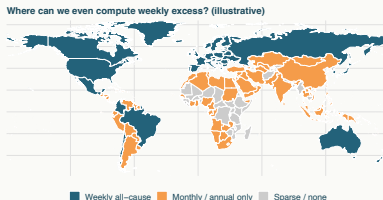
The *empirical* estimand is constrained by what data exist. The *theoretical* estimand should drive what data we prioritize.

¹⁶ Hill, 1987; Bennett and Horiuchi, 1981, 1984

Measurement infrastructure shapes inference

- **Survey-based / indirect estimation** — Demographic and Health Surveys, sibling-survival histories
- **Administrative proxies** — burial records, hospital deaths, insurance claims; useful in low-VR settings, each with its own selection mechanism.
- **Conflict / disaster** — retrospective mortality surveys¹⁷, satellite imagery of grave sites: the “observed” side is itself estimated.

COVID lesson: data-quality heterogeneity was enormous — ranking countries on excess is partly ranking vital-registration completeness.¹⁸



¹⁷ Checchi and Roberts, 2005; Burnham et al., 2006

¹⁸ Karlinsky, 2022, 2024

For any *rate* estimand, we need P_{it} .

- Mid-year population estimates rely on a census plus migration assumptions.
- Census lag (often 5–10 years from latest enumeration) propagates into every rate.
- Pandemic-era *migration disruption* (urban exodus, halted international migration) breaks the assumptions standard projections rely on.

Consequences:

- Age-specific rates are particularly fragile in age bands where migration is largest (working-age adults).
- Country-level rate comparisons can be dominated by denominator differences.
- In settings with unstable populations, the *count* estimand may be more defensible than the rate.

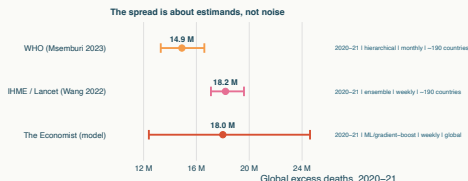
Tradeoff: rates are scientifically cleaner; counts are sometimes empirically cleaner.

Synthesis

Methodological divergence among major estimates

- WHO, IHME, *The Economist*, WMD (Karlinsky–Kobak), and others.
- Differences across these estimates are *not* primarily statistical noise.
- Differences trace to *different estimands* and *different assumptions*.¹⁹

The disagreement is interpretable: each group answered a slightly different question, with slightly different data, under slightly different assumptions.



¹⁹ Msemburi et al., 2023; Wang et al., 2022a; Karlinsky, 2024; Wang et al., 2022b; Economist and Solstad, 2021

Before fitting any model, write down:

1. What *scientific question* matters here — burden, risk, disruption, or causal effect?
2. What *counterfactual world* is plausible and relevant?
3. What *data* are credible, and at what scale?
4. Which *theoretical estimand* aligns with the question?

Disagreement at step (4) is substantive disagreement about the question — not a methodological flaw to be hidden behind sensitivity tables.

These four questions live *outside* the model. They should be answerable before any code is written.

Excess mortality is not one thing.

It is a family of *model-based counterfactual contrasts*:

- not simply observed minus expected,
- not purely descriptive,
- not directly observed.

Different choices imply different answers *by design*, not by error.

When we estimate excess mortality, the first question is not “what model should we use?”, but “what *exactly* are we trying to estimate?”

For the 1EME workshop: this is what the six estimands can make visible.

- Jon Wakefield (UW Biostatistics) — co-author and advisor.
- Collaborators: WHO and group (Jonas Schöley, Ariel Karlinsky, Enrique Acosta, Nelly Biondi, William Msemburi, Haidong Wang)

Thank you. Questions?

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